Supplemental Material

Data S1

Search Strategy

We performed a systematic search of PubMed, Embase, Cochrane Library and Web of Science from 01/2010 to 04/2021. The search was restricted to 'Article'. There was no language restriction.

PubMed search strategy:

((((((((((Ventricular Dysfunction, Right) OR Right Ventricular Dysfunction) OR Dysfunction, Right Ventricular) OR Dysfunctions, Right Ventricular) OR Right Ventricular Dysfunctions) OR Ventricular Dysfunctions, Right) OR right ventricular failure)) OR ((((((((((((((Heart Failure) OR Cardiac Failure) OR Heart Decompensation) OR Decompensation, Heart) OR Heart Failure, Right-Sided) OR Heart Failure, Right Sided) OR Right-Sided Heart Failure) OR Right Sided Heart Failure) OR Myocardial Failure) OR Congestive Heart Failure) OR Heart Failure, Congestive) OR Heart Failure, Left-Sided) OR Heart Failure, Left Sided) OR Left-Sided Heart Failure) OR Left Sided Heart Failure)))) AND ((((((((((sacubitril-valsartan) OR sacubitril valsartan sodium hydrate) OR sacubitril valsartan drug combination) OR sacubitril) OR entresto) OR LCZ 696) OR LCZ696) OR LCZ-696) OR Sacubitril/valsartan) OR Angiotensin Receptor-Neprilysin Inhibitor) OR Sacubitril/Valsartan)

Filters: from 2010 – 2021

Total 1332

Embase search strategy:

('heart right ventricle function'/exp OR 'heart right ventricle function':ab,ti OR 'right ventricular dysfunction':ab,ti OR 'dysfunction, right ventricular':ab,ti OR 'dysfunctions, right ventricular':ab,ti OR 'right ventricular dysfunctions':ab,ti OR 'ventricular dysfunctions, right':ab,ti OR 'right ventricular failure':ab,ti OR 'heart failure'/exp OR 'cardiac failure':ab,ti OR 'heart decompensation':ab,ti OR 'decompensation, heart':ab,ti OR 'heart failure, right-sided':ab,ti OR 'heart failure, right sided':ab,ti OR 'right-sided heart failure':ab,ti OR 'right sided heart failure':ab,ti OR 'myocardial failure':ab,ti OR 'congestive heart failure':ab,ti OR 'heart failure, congestive':ab,ti OR 'heart failure, left-sided':ab,ti OR 'heart failure, left sided':ab,ti OR 'left-sided heart failure':ab,ti OR 'left sided heart failure':ab,ti) AND ('sacubitril plus valsartan'/exp OR 'sacubitril-valsartan':ab,ti OR 'sacubitril valsartan sodium hydrate':ab,ti OR 'sacubitril valsartan drug combination':ab,ti OR 'sacubitril':ab,ti OR 'entresto':ab,ti OR 'lcz 696':ab,ti OR 'lcz696':ab,ti OR 'lcz-696':ab,ti OR 'angiotensin receptor-neprilysin inhibitor':ab,ti OR 'sacubitril/valsartan':ab,ti) AND [2010-2021]/py AND [article]/lim

Total 834

The Cochrane Library search strategy:

#1 MeSH descriptor: [Heart Failure] explode all trees

#2 (cardiac failure) OR (heart decompensation) OR (decompensation, heart) OR (heart failure, right-sided) OR (heart failure, right sided) OR (right-sided heart failure) OR (right sided heart failure) OR (myocardial failure) OR (congestive heart failure) OR (heart failure, congestive)

OR (heart failure, left-sided) OR (heart failure, left sided) OR (left-sided heart failure) OR (left sided heart failure)

#3 #1 OR #2

#4 MeSH descriptor: [Ventricular Dysfunction, Right] explode all trees

#5 (heart right ventricle function) OR (right ventricular dysfunction) OR (dysfunction, right ventricular) OR (dysfunctions, right ventricular) OR (right ventricular failure)

#6 #4 OR #5

#7 #3 OR #6

#8 (sacubitril plus valsartan) OR (sacubitril-valsartan) OR (sacubitril valsartan sodium hydrate) OR (sacubitril valsartan drug combination) OR (sacubitril) OR (entresto) OR (lcz 696) OR (lcz696) OR (lcz-696) OR (angiotensin receptor-neprilysin inhibitor)

#9 #7 AND #8

2010.01-2021.04；Trial

Total 245

Web of Science search strategy:

#1 (TS = (sacubitril-valsartan OR sacubitril valsartan sodium hydrate OR sacubitril valsartan drug combination OR sacubitril OR entresto OR LCZ 696 OR LCZ696 OR LCZ-696 OR Sacubitril/valsartan OR Angiotensin Receptor-Neprilysin Inhibitor OR Sacubitril/Valsartan))

#2 (TS=(Heart Failure or Cardiac Failure or Heart Decompensation or Decompensation, Heart or Heart Failure, Right-Sided or Heart Failure, Right Sided or Right Sided Heart Failure or Right Sided Heart Failure or Myocardial Failure or Congestive Heart Failure or Heart Failure, Congestive or Heart Failure, Left-Sided or Heart Failure, Left Sided or Left-Sided Heart Failure or Left Sided Heart Failure or Ventricular Dysfunction, Right or Right Ventricular Dysfunction or Dysfunction, Right Ventricular or Dysfunctions, Right Ventricular or Right Ventricular Dysfunctions or Ventricular Dysfunctions, Right or right ventricular failure))

#3 Restrictive conditions: Time period: 2010-2021; Article types: ARTICLE

#4 #1 and #2 AND #3

Total 1259

Table S1. Quality assessment of included studies according to the Newcastle-Ottawa Scale (NOS).

First Author (Year)	Selection	Comparability	Outcome	Score/Maximum
Eleni S. (2018)[24]	★★★★	★★	★★★	9/9
Francesco (2020)[22]	★★★	★	★★★	7/9
Geoffrey (2019)[15]	★★★	★	★★★	7/9
Michele (2020)[13]	★★★	★	★★★	7/9
Gregor (2020)[25]	★★★	★	★★★	7/9
Simone (2020)[23]	★★★	★	★★★	7/9
Alessandra (2020)[27]	★★	★	★★★	6/9
Mustafa (2020)[28]	★★	★	★★★	6/9
Matteo (2020)[26]	★★	★	★★★	6/9
Daniele (2020)[14]	★★	★	★★★	6/9

A maximum of 4 stars for selection, 2 for comparability and 3 for the outcome.

Table S2. Changes of RV system indices following the treatment of S/V.

First Author (Year)	TAPSE (mm)		S' (cm/sec)		sPAP (mmHg)		mPAP (mmHg)	
	Pre-S/V	Post-S/V	Pre-S/V	Post-S/V	Pre-S/V	Post-S/V	Pre-S/V	Post-S/V
Eleni S. (2018)[24]	14.8 ±2.8	15.7 ±2.2	10.2 ± 2.8	11.3 ± 2.1	NA	NA	NA	NA
Francesco (2020)[22]	16.5 ± 4.6	16.3 ± 3.7	NA	NA	45.3 ± 12.3	40.6 ± 11.1	32.9 ± 9.9	29.7 ± 9.5
Geoffrey (2019)[15]	18±4	18±4	NA	NA	39 ± 10	32±8	NA	NA
Michele (2020)[13]	16.5 ± 4.0	17.8 ± 3.9	NA	NA	34.7 ± 12.5	31.0 ± 12.8	NA	NA
Gregor (2020)[25]	17 ± 5	21 ± 6	NA	NA	NA	NA	NA	NA
Simone (2020)[23]	17 ± 4	17 ± 7	NA	NA	NA	NA	NA	NA
Alessandra (2020)[27]	NA	NA	NA	NA	48.6 ±22.8	39.3±18.1	NA	NA
Mustafa (2020)[28]	18.5 ± 2.6	19.7 ± 2.4	NA	NA	NA	NA	33.9 ± 3.7	30.1 ± 2.9
Matteo (2020)[26]	NA	NA	NA	NA	38.8 ± 10.7	32.4 ± 12.2	NA	NA
Daniele (2020)[14]	18.76 ± 3.7	19.3 ± 3.2	10.4 ± 3.2	11.2 ± 2.9	38.3 ± 15.7	29.1 ± 14.8	24.1 + 12.6	22.7 ± 10.9

TAPSE, tricuspid annular plane systolic excursion; S', tricuspid annular peak systolic velocity; sPAP, systolic pulmonary arterial pressure; mPAP, mean pulmonary arterial pressure; S/V, sacubitril/valsartan; NA, not applicable.

Table S3. Changes of left heart indices and NT-proBNP following the treatment of S/V.

First Author (Year)	LVEF (%)		LVEDV (mL)		NT-proBNP (ng/dL)	
	Pre-S/V	Post-S/V	Pre-S/V	Post-S/V	Pre-S/V	Post-S/V
Eleni S. (2018)[24]	30.23 ±3.7	30.23 ± 3.7	NA	NA	NA	NA
Francesco (2020)[22]	23.5 ± 5.8	24.4 ± 6.3	NA	NA	4943.0 ± 5326.8	2257.9 ± 3413.1
Geoffrey (2019)[15]	32.6±5	36 ± 6	172±49	166 ± 58	NA	NA
Michele (2020)[13]	34.0 ± 9.2	39.5 ± 9.8	177.3 ± 71.1	174.4 ± 70.1	3049.1 ± 5775.1	2305.2 ± 6768.4
Gregor (2020)[25]	29.7% ± 8	36.5% ± 9	NA	NA	NA	NA
Simone (2020)[23]	28 ± 8	34 ± 12	178.36 ± 64.15	163.08 ± 85.22	NA	NA
Alessandra (2020)[27]	29.5±5.2	33±7	NA	NA	1819 ±798	1112 ±886
Mustafa (2020)[28]	30.6 ± 2.9	31.7 ± 2.5	NA	NA	1627.3 ± 826.3	895.7 ± 402.5
Matteo (2020)[26]	33.8 ± 6.8	50.3±9.8	169.7 ± 51.4	129.7 ± 49.4	NA	NA
Daniele (2020)[14]	28.9 ± 6.4	31.5 ± 6.2	237.2 ± 87.6	213.3 ± 64.8	NA	NA

LVEF, left ventricular ejection fraction; LVEDV, left ventricular end-diastolic volume; NT-proBNP, N terminal pro B type natriuretic peptide; S/V, sacubitril/valsartan; NA, not applicable.

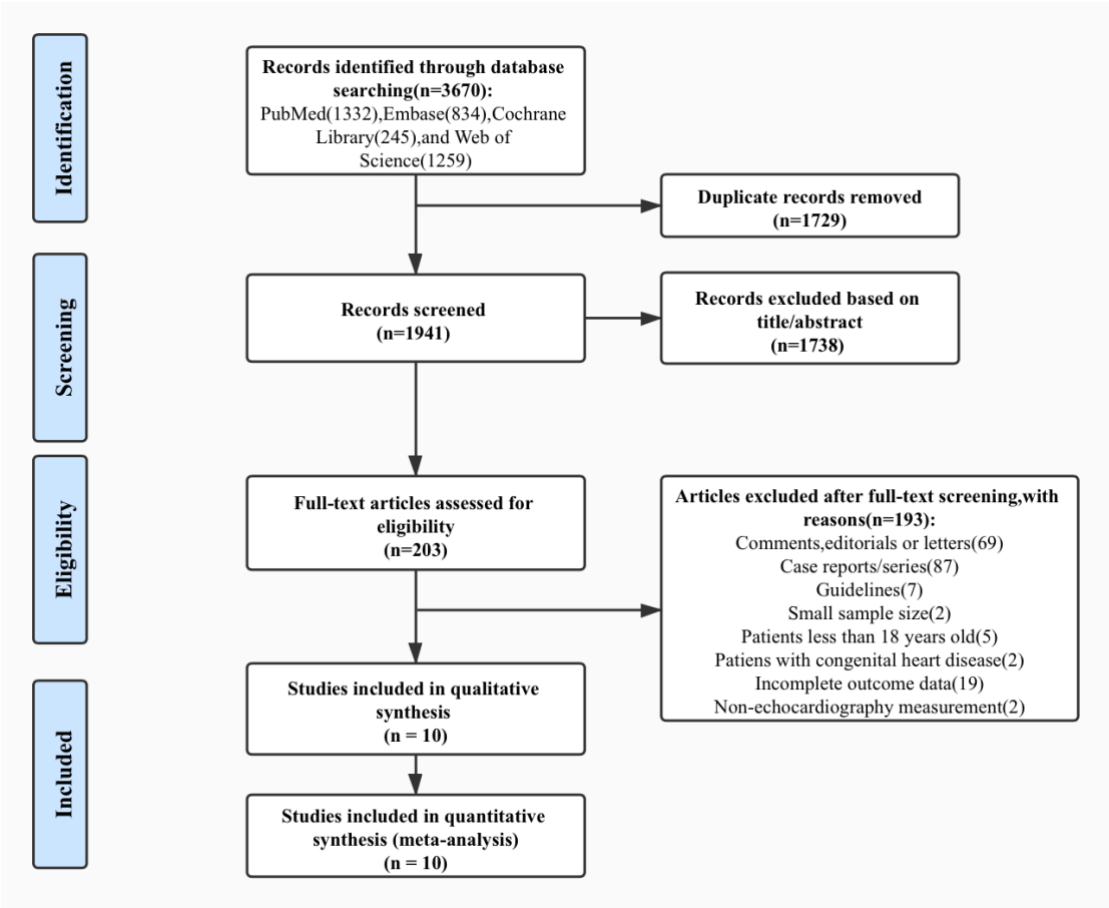


Figure S1. Preferred Reporting Items of Systematic Review and Meta-Analysis (PRISMA) flowchart of study selection.

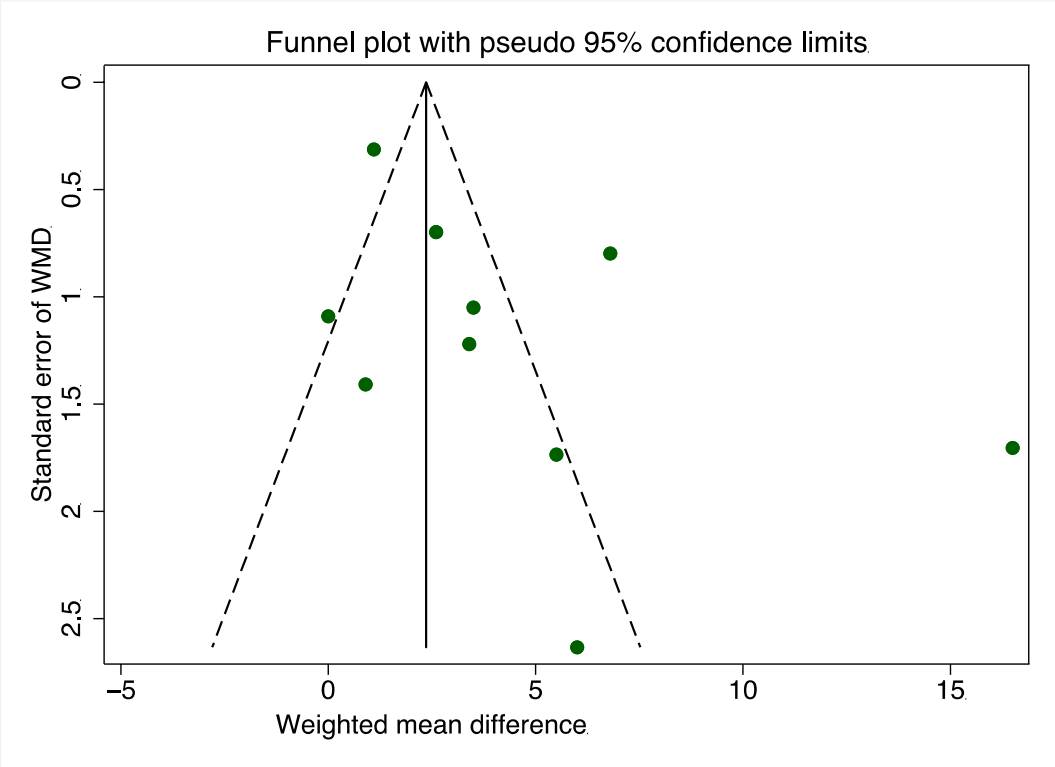


Figure S2. Funnel plots estimating publication bias for LVEF.

LVEF, left ventricular ejection fraction; WMD, weighted mean differences.

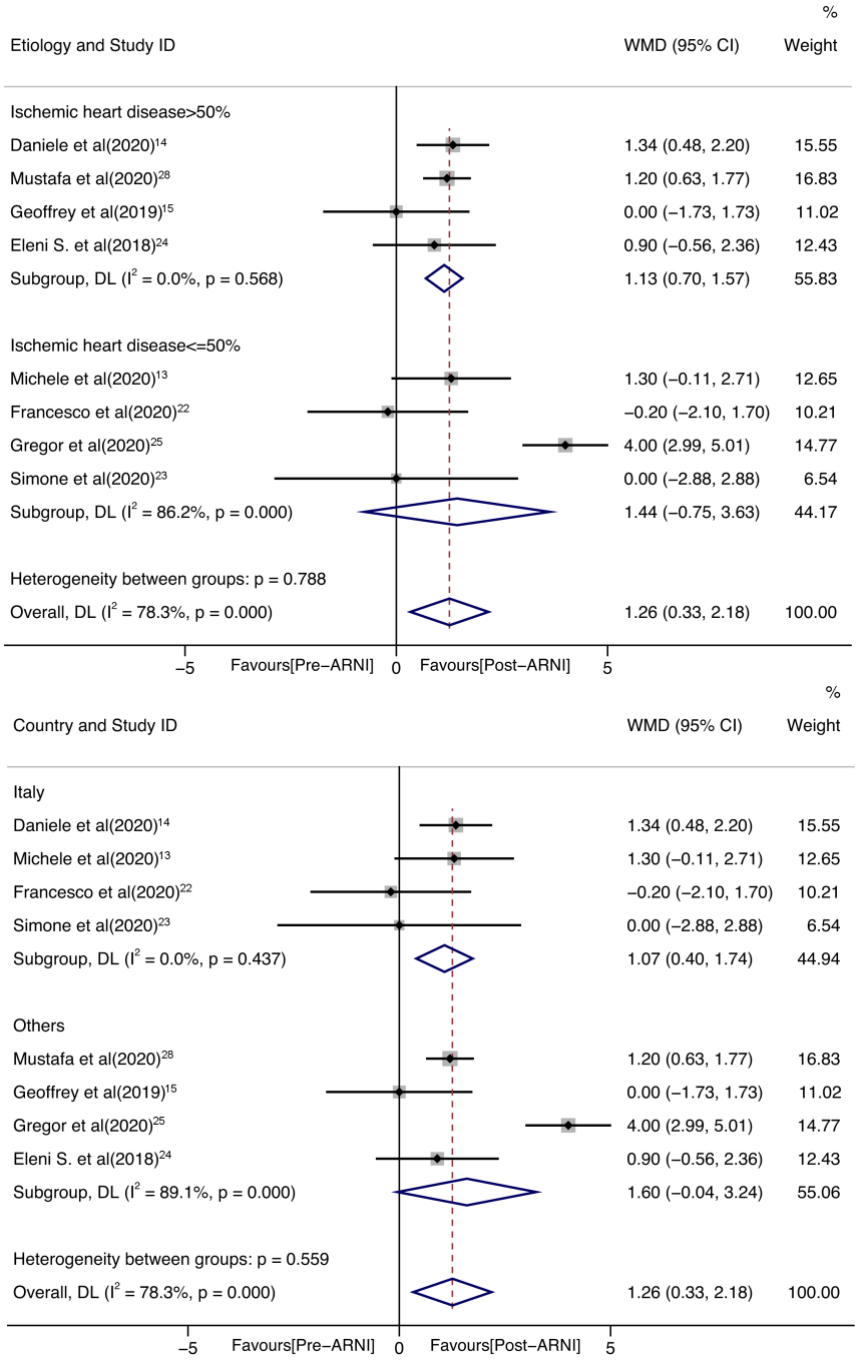


Figure S3. Subgroup analysis of effects of S/V on TAPSE according to HF etiology and country.

S/V, sacubitril/valsartan; TAPSE, tricuspid annular plane systolic excursion; WMD, weighted mean differences; CI, confidence interval; ARNI, angiotensin receptor neprilysin inhibitor; HF, heart failure.

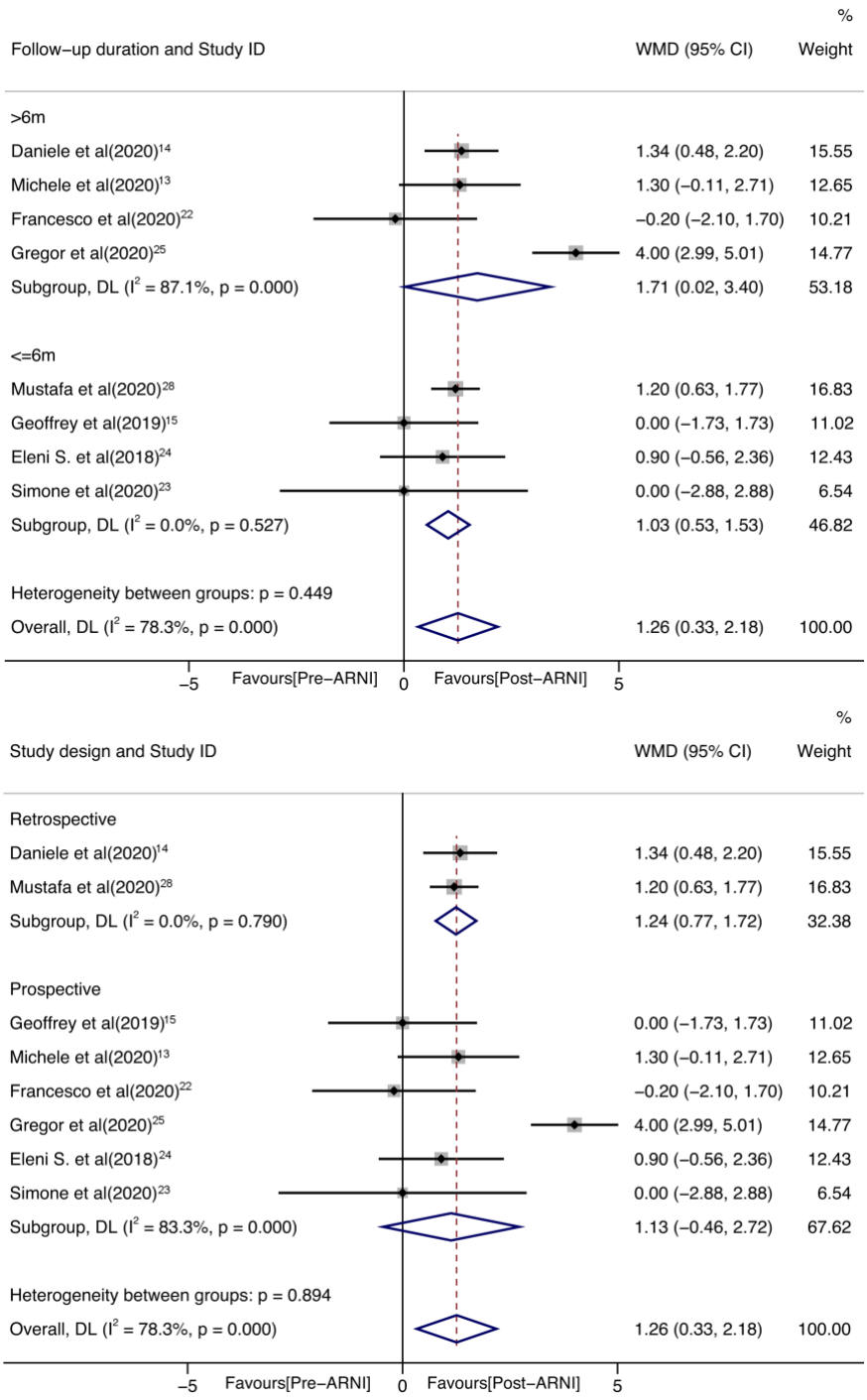


Figure S4. Subgroup analysis of effects of S/V on TAPSE according to follow-up duration and study design.

S/V, sacubitril/valsartan; TAPSE, tricuspid annular plane systolic excursion; WMD, weighted mean differences; CI, confidence interval; ARNI, angiotensin receptor neprilysin inhibitor; HF, heart failure.

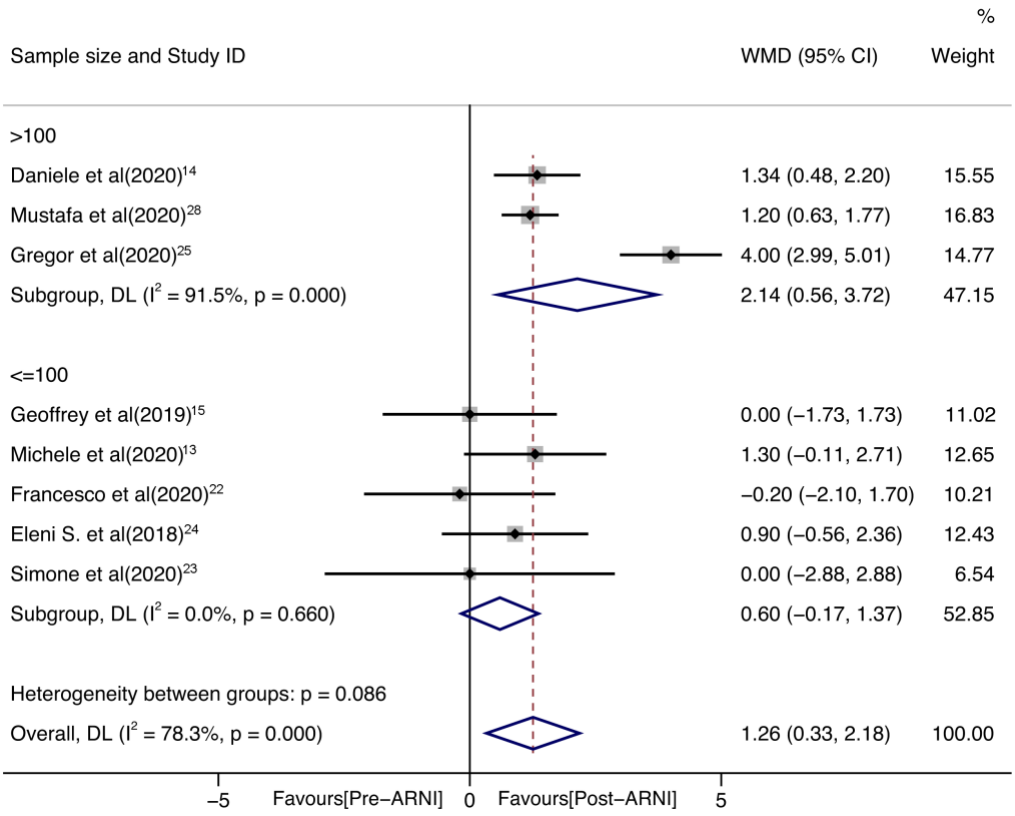


Figure S5. Subgroup analysis of effects of S/V on TAPSE according to sample size.

S/V, sacubitril/valsartan; TAPSE, tricuspid annular plane systolic excursion; WMD, weighted mean differences; CI, confidence interval; ARNI, angiotensin receptor neprilysin inhibitor; HF, heart failure.

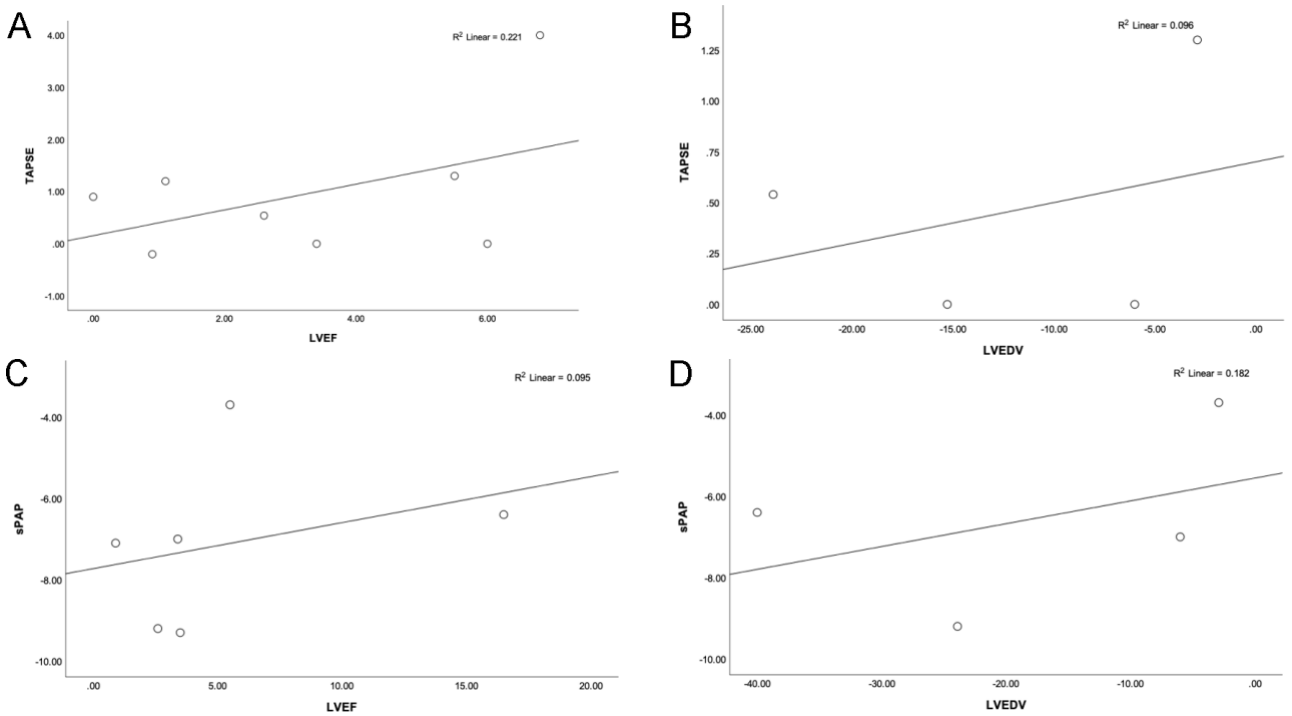


Figure S6. Correlation analysis of RV system and left heart function. (A) TAPSE and LVEF; (B) TAPSE and LVEDV; (C) sPAP and LVEF; (D) sPAP and LVEDV.

TAPSE, tricuspid annular plane systolic excursion; sPAP, systolic pulmonary arterial pressure; LVEF, left ventricular ejection fraction; LVEDV, left ventricular end-diastolic volume; RV, right ventricular.

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page2,3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4,5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Data S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 8, 9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 8
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 8
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 8
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 8
RESULTS			

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 10, Table S1
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 10, Figure 1,2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 10
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page 10, Figure 1,2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Page 11
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 10
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 12
	23b	Discuss any limitations of the evidence included in the review.	Page 16
	23c	Discuss any limitations of the review processes used.	Page 16
	23d	Discuss implications of the results for practice, policy, and future research.	Page 16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 5
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 17
Competing interests	26	Declare any competing interests of review authors.	Page 17
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Supplemental material

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

MOOSE checklist.

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	4,5
2	Hypothesis statement	5
3	Description of study outcome(s)	5
4	Type of exposure or intervention used	5
5	Type of study designs used	5
6	Study population	5
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	N/A
8	Search strategy, including time period included in the synthesis and keywords	6, Data S1
9	Effort to include all available studies, including contact with authors	N/A
10	Databases and registries searched	6
11	Search software used, name and version, including special features used (eg, explosion)	7
12	Use of hand searching (eg, reference lists of obtained articles)	6
13	List of citations located and those excluded, including justification	6, Figure S1
14	Method of addressing articles published in languages other than English	6
15	Method of handling abstracts and unpublished studies	N/A
16	Description of any contact with authors	N/A
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	6
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	6, 7
19	Documentation of how data were classified and coded (eg, multiple raters, blinding, and interrater reliability)	7
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	N/A
21	Assessment of study quality, including blinding of quality assessors; stratification or regression on possible predictors of study results	7
22	Assessment of heterogeneity	8

23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	8
24	Provision of appropriate tables and graphics	2 supplemental checklists, Data S1
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Figure1, Figure2
26	Table giving descriptive information for each study included	Table 1
27	Results of sensitivity testing (eg, subgroup analysis)	11, Table 3
28	Indication of statistical uncertainty of findings	11,12, Table 3, Figure 3
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	10, Table S1, Figure S2
30	Justification for exclusion (eg, exclusion of non–English-language citations)	N/A
31	Assessment of quality of included studies	10
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	13-15
33	Generalisation of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	16
34	Guidelines for future research	16
35	Disclosure of funding source	17

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta- analysis of Observational Studies in Epidemiology. A Proposal for Reporting. JAMA. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.